Acids

Mineral (inorganic) acids
- Hydrochloric acid – HCl
- Hydrofluoric acid – HF
- Hydrobromic acid – HBr
- Hydroiodic acid – HI
- Sulfuric acid – H_2SO_4
- Sulfurous acid – H_2SO_3
- Nitric acid – HNO_3
- Nitrous acid – HNO_2
- Phosphoric acid – H_3PO_4
- Phosphorous acid – H_3PO_3
- Hypophosphorous acid – H_3PO_2
- Carbonic acid – H_2CO_3
- Boric acid – H_3BO_3
- Silicic acid – H_4SiO_4
- Chloric acid – $HClO_3$
- Perchloric acid – $HClO_4$
- Chlorous acid – $HClO_2$
- Hypochlorous acid – $HClO$
- Bromic acid – $HBrO_3$
- Perbromic acid – $HBrO_4$
- Iodic acid – HIO_3
- Periodic acid – HIO_4
- Chromic acid – H_2CrO_4
- Dichromic acid – $H_2Cr_2O_7$
- Permanganic acid – $HMnO_4$
- Hydrocyanic acid – HCN
- Hydrazoic acid – HN_3
- Hydrosulfuric acid (hydrogen sulfide) – H_2S
- Thiosulfuric acid – $H_2S_2O_3$
- Peroxymonosulfuric acid (Caro's acid) – H_2SO_5
- Peroxydisulfuric acid – $H_2S_2O_8$
- Pyrophosphoric acid – $H_4P_2O_7$
- Metaphosphoric acid – HPO_3
- Arsenic acid – H_3AsO_4
- Arsenous acid – H_3AsO_3
- Selenic acid – H_2SeO_4
- Selenous acid – H_2SeO_3
- Telluric acid – H_6TeO_6
- Hydroselenic acid – H_2Se
- Hydrotelluric acid – H_2Te
- Molybdic acid – H_2MoO_4
- Tungstic acid – H_2WO_4
- Fluorosilicic acid – H_2SiF_6

- Fluoroboric acid – HBF_4
- Hexafluorophosphoric acid – HPF_6
- Hexafluoroantimonic acid – $HSbF_6$
- Chloroauric acid – $HAuCl_4$
- Hexachloroplatinic acid – H_2PtCl_6
- Thiocyanic acid – HSCN
- Cyanic acid – HOCN
- Isocyanic acid – HNCO
- Fulminic acid – HCNO
- Sulfamic acid – H_3NSO_3
- Nitrosylsulfuric acid – $HNSO_5$
- Xenic acid – H_2XeO_4
- Orthoperiodic acid – H_5IO_6

Superacids
- Fluoroantimonic acid – $HSbF_6$
- Magic acid – $FSO_3H{\cdot}SbF_5$
- Fluorosulfuric acid – HSO_3F
- Triflic acid – CF_3SO_3H
- Carborane acid – $H(CHB_{11}Cl_{11})$

Simple carboxylic acids
- Formic acid – HCOOH
- Acetic acid – CH_3COOH
- Propionic acid – C_2H_5COOH
- Butyric acid – C_3H_7COOH
- Valeric acid – C_4H_9COOH
- Caproic acid – $C_5H_{11}COOH$
- Enanthic acid – $C_6H_{13}COOH$
- Caprylic acid – $C_7H_{15}COOH$
- Pelargonic acid – $C_8H_{17}COOH$
- Capric acid – $C_9H_{19}COOH$
- Isobutyric acid – $(CH_3)_2CHCOOH$
- Pivalic acid – $(CH_3)_3CCOOH$

Halogenated acetic acids
- Chloroacetic acid – $ClCH_2COOH$
- Dichloroacetic acid – $Cl_2CHCOOH$
- Trichloroacetic acid – CCl_3COOH
- Trifluoroacetic acid – CF_3COOH
- Bromoacetic acid – $BrCH_2COOH$
- Iodoacetic acid – ICH_2COOH

Dicarboxylic acids
- Oxalic acid – HOOC–COOH
- Malonic acid – $CH_2(COOH)_2$
- Succinic acid – $(CH_2)_2(COOH)_2$
- Glutaric acid – $(CH_2)_3(COOH)_2$

- Adipic acid – $(CH_2)_4(COOH)_2$
- Pimelic acid – $(CH_2)_5(COOH)_2$
- Suberic acid – $(CH_2)_6(COOH)_2$
- Azelaic acid – $(CH_2)_7(COOH)_2$
- Sebacic acid – $(CH_2)_8(COOH)_2$
- Maleic acid – cis-HOOCCH=CHCOOH
- Fumaric acid – trans-HOOCCH=CHCOOH
- Phthalic acid – $C_6H_4(COOH)_2$ (ortho)
- Isophthalic acid – $C_6H_4(COOH)_2$ (meta)
- Terephthalic acid – $C_6H_4(COOH)_2$ (para)

Hydroxy and keto acids
- Glycolic acid – $HOCH_2COOH$
- Lactic acid – $CH_3CH(OH)COOH$
- Citric acid – $C_6H_8O_7$
- Isocitric acid – $C_6H_8O_7$
- Malic acid – $C_4H_6O_5$
- Tartaric acid – $C_4H_6O_6$
- Mandelic acid – $C_6H_5CH(OH)COOH$
- Gluconic acid – $C_6H_{12}O_7$
- Glucuronic acid – $C_6H_{10}O_7$
- Glyceric acid – $C_3H_6O_4$
- Pyruvic acid – $CH_3COCOOH$
- Oxaloacetic acid – $C_4H_4O_5$
- α-Ketoglutaric acid – $C_5H_6O_5$
- Acetoacetic acid – CH_3COCH_2COOH
- Levulinic acid – $CH_3CO(CH_2)_2COOH$
- β-Hydroxybutyric acid – $C_4H_8O_3$
- Mevalonic acid – $C_6H_{12}O_4$
- Quinic acid – $C_7H_{12}O_6$
- Shikimic acid – $C_7H_{10}O_5$
- Tropic acid – $C_9H_{10}O_3$
- Hydroxycitric acid – $C_6H_8O_8$

Aromatic acids
- Benzoic acid – C_6H_5COOH
- Salicylic acid – $C_7H_6O_3$
- Acetylsalicylic acid (aspirin) – $C_9H_8O_4$
- Gallic acid – $C_7H_6O_5$
- Cinnamic acid – $C_9H_8O_2$
- Caffeic acid – $C_9H_8O_4$
- Ferulic acid – $C_{10}H_{10}O_4$
- p-Coumaric acid – $C_9H_8O_3$
- Vanillic acid – $C_8H_8O_4$
- Protocatechuic acid – $C_7H_6O_4$
- Syringic acid – $C_9H_{10}O_5$
- Anthranilic acid – $C_7H_7NO_2$
- Phenylacetic acid – $C_6H_5CH_2COOH$

- Nicotinic acid (niacin) – $C_6H_5NO_2$
- Picric acid – $C_6H_3N_3O_7$
- Carbolic acid (phenol) – C_6H_5OH
- Tannic acid – $C_{76}H_{52}O_{46}$
- Ellagic acid – $C_{14}H_6O_8$
- Chlorogenic acid – $C_{16}H_{18}O_9$
- Rosmarinic acid – $C_{18}H_{16}O_8$

Saturated fatty acids
- Lauric acid – $C_{11}H_{23}COOH$
- Myristic acid – $C_{13}H_{27}COOH$
- Palmitic acid – $C_{15}H_{31}COOH$
- Stearic acid – $C_{17}H_{35}COOH$
- Arachidic acid – $C_{19}H_{39}COOH$
- Behenic acid – $C_{21}H_{43}COOH$
- Lignoceric acid – $C_{23}H_{47}COOH$
- Cerotic acid – $C_{25}H_{51}COOH$

Unsaturated fatty acids
- Oleic acid – $C_{17}H_{33}COOH$
- Linoleic acid – $C_{17}H_{31}COOH$
- α-Linolenic acid – $C_{17}H_{29}COOH$
- γ-Linolenic acid – $C_{17}H_{29}COOH$
- Arachidonic acid – $C_{19}H_{31}COOH$
- Eicosapentaenoic acid (EPA) – $C_{19}H_{29}COOH$
- Docosahexaenoic acid (DHA) – $C_{21}H_{31}COOH$
- Palmitoleic acid – $C_{15}H_{29}COOH$
- Erucic acid – $C_{21}H_{41}COOH$
- Nervonic acid – $C_{23}H_{4}$$500COOH$
- Ricinoleic acid – $C_{17}H_{32}(OH)COOH$
- Sorbic acid – C_5H_7COOH
- Acrylic acid – $CH_2=CHCOOH$
- Methacrylic acid – $CH_2=C(CH_3)COOH$
- Crotonic acid – $CH_3CH=CHCOOH$

Amino acids (standard 20)
- Glycine – $C_2H_5NO_2$
- Alanine – $C_3H_7NO_2$
- Valine – $C_5H_{11}NO_2$
- Leucine – $C_6H_{13}NO_2$
- Isoleucine – $C_6H_{13}NO_2$
- Proline – $C_5H_9NO_2$
- Phenylalanine – $C_9H_{11}NO_2$
- Tryptophan – $C_{11}H_{12}N_2O_2$
- Methionine – $C_5H_{11}NO_2S$
- Serine – $C_3H_7NO_3$
- Threonine – $C_4H_9NO_3$
- Cysteine – $C_3H_7NO_2S$

- Tyrosine – $C_9H_{11}NO_3$
- Asparagine – $C_4H_8N_2O_3$
- Glutamine – $C_5H_{10}N_2O_3$
- Aspartic acid – $C_4H_7NO_4$
- Glutamic acid – $C_5H_9NO_4$
- Lysine – $C_6H_{14}N_2O_2$
- Arginine – $C_6H_{14}N_4O_2$
- Histidine – $C_6H_9N_3O_2$

Other amino acids
- Selenocysteine – $C_3H_7NO_2Se$
- Pyrrolysine – $C_{12}H_{21}N_3O_3$
- Ornithine – $C_5H_{12}N_2O_2$
- Citrulline – $C_6H_{13}N_3O_3$
- Taurine – $C_2H_7NO_3S$
- GABA – $C_4H_9NO_2$
- β-Alanine – $C_3H_7NO_2$

Nucleic acids
- Deoxyribonucleic acid (DNA) – $(C_5H_7O_6PR)$□ polymer
- Ribonucleic acid (RNA) – $(C_5H_7O_7PR)$□ polymer
- Adenosine triphosphate (ATP) – $C_{10}H_{16}N_5O_{13}P_3$

Sulfonic and organic sulfur acids
- Methanesulfonic acid – CH_3SO_3H
- Benzenesulfonic acid – $C_6H_5SO_3H$
- p-Toluenesulfonic acid – $CH_3C_6H_4SO_3H$
- Sulfanilic acid – $H_2NC_6H_4SO_3H$
- Camphorsulfonic acid – $C_{10}H_{16}O_4S$
- Lipoic acid – $C_8H_{14}O_2S_2$
- Cysteic acid – $C_3H_7NO_5S$

Vitamins and biological acids
- Ascorbic acid (vitamin C) – $C_6H_8O_6$
- Folic acid – $C_{19}H_{19}N_7O_6$
- Pantothenic acid – $C_9H_{17}NO_5$
- Retinoic acid – $C_{20}H_{28}O_2$
- Cholic acid – $C_{24}H_{40}O_5$
- Deoxycholic acid – $C_{24}H_{40}O_4$
- Chenodeoxycholic acid – $C_{24}H_{40}O_4$
- Hyaluronic acid – $(C_{14}H_{21}NO_{11})$□
- Uric acid – $C_5H_4N_4O_3$
- Orotic acid – $C_5H_4N_2O_4$
- Sialic acid – $C_{11}H_{19}NO_9$
- Abscisic acid – $C_{15}H_{20}O_4$
- Gibberellic acid – $C_{19}H_{22}O_6$
- Indole-3-acetic acid – $C_{10}H_9NO_2$
- Jasmonic acid – $C_{12}H_{18}O_3$

- Humic acid – variable composition
- Fulvic acid – variable composition

Other notable organic acids
- Barbituric acid – $C_4H_4N_2O_3$
- Cyanuric acid – $C_3H_3N_3O_3$
- Isocyanuric acid – $C_3H_3N_3O_3$
- Squaric acid – $C_4H_2O_4$
- Ethylenediaminetetraacetic acid (EDTA) – $C_{10}H_{16}N_2O_8$
- Nitrilotriacetic acid – $C_6H_9NO_6$
- Phytic acid – $C_6H_{18}O_{24}P_6$
- Domoic acid – $C_{15}H_{21}NO_6$
- Kainic acid – $C_{10}H_{15}NO_4$
- Usnic acid – $C_{18}H_{16}O_7$
- Carminic acid – $C_{22}H_{20}O_{13}$
- Glycyrrhizic acid – $C_{42}H_{62}O_{16}$
- Valproic acid – $C_8H_{16}O_2$
- Ibuprofen – $C_{13}H_{18}O_2$
- Mefenamic acid – $C_{15}H_{15}NO_2$
- Tranexamic acid – $C_8H_{15}NO_2$
- Boronic acid (general) – $RB(OH)_2$
- Phenylboronic acid – $C_6H_5B(OH)_2$
- Phosphonic acid – H_3PO_3
- Glyphosate – $C_3H_8NO_5P$
- Peracetic acid – CH_3CO_3H
- Performic acid – HCO_3H
- Perbenzoic acid – $C_6H_5CO_3H$

Bases

Alkali metal hydroxides
- Lithium hydroxide – LiOH
- Sodium hydroxide – NaOH
- Potassium hydroxide – KOH
- Rubidium hydroxide – RbOH
- Caesium hydroxide – CsOH
- Francium hydroxide – FrOH

Alkaline earth hydroxides
- Beryllium hydroxide – $Be(OH)_2$
- Magnesium hydroxide – $Mg(OH)_2$
- Calcium hydroxide – $Ca(OH)_2$
- Strontium hydroxide – $Sr(OH)_2$
- Barium hydroxide – $Ba(OH)_2$
- Radium hydroxide – $Ra(OH)_2$

Other metal hydroxides
- Aluminium hydroxide – $Al(OH)_3$
- Zinc hydroxide – $Zn(OH)_2$
- Iron(II) hydroxide – $Fe(OH)_2$
- Iron(III) hydroxide – $Fe(OH)_3$
- Copper(II) hydroxide – $Cu(OH)_2$
- Nickel(II) hydroxide – $Ni(OH)_2$
- Cobalt(II) hydroxide – $Co(OH)_2$
- Manganese(II) hydroxide – $Mn(OH)_2$
- Chromium(III) hydroxide – $Cr(OH)_3$
- Lead(II) hydroxide – $Pb(OH)_2$
- Tin(II) hydroxide – $Sn(OH)_2$
- Silver hydroxide – $AgOH$
- Cadmium hydroxide – $Cd(OH)_2$
- Thallium(I) hydroxide – $TlOH$
- Gallium hydroxide – $Ga(OH)_3$
- Indium hydroxide – $In(OH)_3$
- Scandium hydroxide – $Sc(OH)_3$
- Yttrium hydroxide – $Y(OH)_3$
- Lanthanum hydroxide – $La(OH)_3$
- Cerium(III) hydroxide – $Ce(OH)_3$
- Bismuth hydroxide – $Bi(OH)_3$
- Zirconium hydroxide – $Zr(OH)_4$
- Titanium hydroxide – $Ti(OH)_4$

Ammonia and related
- Ammonia – NH_3
- Ammonium hydroxide – NH_4OH
- Hydrazine – N_2H_4
- Hydroxylamine – NH_2OH
- Tetramethylammonium hydroxide – $(CH_3)_4NOH$
- Tetraethylammonium hydroxide – $(C_2H_5)_4NOH$
- Tetrabutylammonium hydroxide – $(C_4H_9)_4NOH$
- Choline hydroxide – $[(CH_3)_3NCH_2CH_2OH]OH$

Basic oxides
- Lithium oxide – Li_2O
- Sodium oxide – Na_2O
- Potassium oxide – K_2O
- Rubidium oxide – Rb_2O
- Caesium oxide – Cs_2O
- Magnesium oxide – MgO
- Calcium oxide – CaO
- Strontium oxide – SrO
- Barium oxide – BaO
- Zinc oxide – ZnO
- Aluminium oxide – Al_2O_3
- Copper(II) oxide – CuO

- Iron(III) oxide – Fe_2O_3
- Lead(II) oxide – PbO
- Silver oxide – Ag_2O
- Mercury(II) oxide – HgO

Basic carbonates and bicarbonates
- Sodium carbonate – Na_2CO_3
- Potassium carbonate – K_2CO_3
- Lithium carbonate – Li_2CO_3
- Caesium carbonate – Cs_2CO_3
- Calcium carbonate – $CaCO_3$
- Magnesium carbonate – $MgCO_3$
- Barium carbonate – $BaCO_3$
- Ammonium carbonate – $(NH_4)_2CO_3$
- Sodium bicarbonate – $NaHCO_3$
- Potassium bicarbonate – $KHCO_3$

Other basic salts and compounds
- Sodium phosphate – Na_3PO_4
- Potassium phosphate – K_3PO_4
- Sodium acetate – CH_3COONa
- Sodium sulfide – Na_2S
- Sodium cyanide – $NaCN$
- Sodium borate (borax) – $Na_2B_4O_7$
- Sodium silicate – Na_2SiO_3
- Sodium amide – $NaNH_2$
- Potassium amide – KNH_2
- Lithium diisopropylamide – $LiN(C_3H_7)_2$
- Sodium hydride – NaH
- Potassium hydride – KH
- Calcium hydride – CaH_2
- Sodium methoxide – CH_3ONa
- Sodium ethoxide – C_2H_5ONa
- Potassium tert-butoxide – $(CH_3)_3COK$
- n-Butyllithium – C_4H_9Li
- Methyllithium – CH_3Li
- Sodium bis(trimethylsilyl)amide – $NaN(Si(CH_3)_3)_2$

Organic amines
- Methylamine – CH_3NH_2
- Dimethylamine – $(CH_3)_2NH$
- Trimethylamine – $(CH_3)_3N$
- Ethylamine – $C_2H_5NH_2$
- Diethylamine – $(C_2H_5)_2NH$
- Triethylamine – $(C_2H_5)_3N$
- Propylamine – $C_3H_7NH_2$
- Isopropylamine – $(CH_3)_2CHNH_2$
- Butylamine – $C_4H_9NH_2$

- Diisopropylethylamine (Hünig's base) – $(C_3H_7)_2NC_2H_5$
- Ethylenediamine – $H_2NCH_2CH_2NH_2$
- Ethanolamine – $HOCH_2CH_2NH_2$
- Diethanolamine – $(HOCH_2CH_2)_2NH$
- Triethanolamine – $(HOCH_2CH_2)_3N$
- Aniline – $C_6H_5NH_2$
- N,N-Dimethylaniline – $C_6H_5N(CH_3)_2$
- Benzylamine – $C_6H_5CH_2NH_2$
- Cyclohexylamine – $C_6H_{11}NH_2$

Heterocyclic and aromatic bases
- Pyridine – C_5H_5N
- 4-Dimethylaminopyridine (DMAP) – $C_7H_{10}N_2$
- 2,6-Lutidine – C_7H_9N
- Pyrrolidine – C_4H_9N
- Piperidine – $C_5H_{11}N$
- Piperazine – $C_4H_{10}N_2$
- Morpholine – C_4H_9NO
- Imidazole – $C_3H_4N_2$
- Pyrimidine – $C_4H_4N_2$
- Quinoline – C_9H_7N
- Quinuclidine – $C_7H_{13}N$
- DABCO – $C_6H_{12}N_2$
- N-Methylmorpholine – $C_5H_{11}NO$
- N-Methylimidazole – $C_4H_6N_2$

Nucleobases
- Adenine – $C_5H_5N_5$
- Guanine – $C_5H_5N_5O$
- Cytosine – $C_4H_5N_3O$
- Thymine – $C_5H_6N_2O_2$
- Uracil – $C_4H_4N_2O_2$

Amidines, guanidines and organic superbases
- Guanidine – CH_5N_3
- Acetamidine – $C_2H_6N_2$
- DBU – $C_9H_{16}N_2$
- DBN – $C_7H_{12}N_2$
- TBD – $C_7H_{13}N_3$
- MTBD – $C_8H_{15}N_3$
- Proton sponge – $C_{14}H_{18}N_2$
- Phosphazene base P1-t-Bu – $C_{10}H_{28}N_4P$

Alkaloids (naturally occurring bases)
- Caffeine – $C_8H_{10}N_4O_2$
- Nicotine – $C_{10}H_{14}N_2$
- Morphine – $C_{17}H_{19}NO_3$
- Codeine – $C_{18}H_{21}NO_3$

- Quinine – $C_{20}H_{24}N_2O_2$
- Atropine – $C_{17}H_{23}NO_3$
- Cocaine – $C_{17}H_{21}NO_4$
- Strychnine – $C_{21}H_{22}N_2O_2$
- Theobromine – $C_7H_8N_4O_2$
- Ephedrine – $C_{10}H_{15}NO$

Others
- Lime water – $Ca(OH)_2$ (aq)
- Soda lime – NaOH + CaO
- Washing soda – $Na_2CO_3 \cdot 10H_2O$
- Baking soda – $NaHCO_3$
- Milk of magnesia – $Mg(OH)_2$ (suspension)

Salts & Solutions

Chlorides
• Sodium chloride – NaCl
• Potassium chloride – KCl
• Calcium chloride – $CaCl_2$
• Magnesium chloride – $MgCl_2$
• Ammonium chloride – NH_4Cl
• Zinc chloride – $ZnCl_2$
• Iron(III) chloride – $FeCl_3$
• Copper(II) chloride – $CuCl_2$
• Silver chloride – AgCl
• Barium chloride – $BaCl_2$
• Aluminium chloride – $AlCl_3$

Sulfates
• Sodium sulfate – Na_2SO_4
• Potassium sulfate – K_2SO_4
• Calcium sulfate – $CaSO_4$
• Magnesium sulfate – $MgSO_4$
• Copper(II) sulfate – $CuSO_4$
• Iron(II) sulfate – $FeSO_4$
• Zinc sulfate – $ZnSO_4$
• Ammonium sulfate – $(NH_4)_2SO_4$
• Barium sulfate – $BaSO_4$
• Aluminium sulfate – $Al_2(SO_4)_3$

Nitrates
• Sodium nitrate – $NaNO_3$
• Potassium nitrate – KNO_3
• Silver nitrate – $AgNO_3$

• Calcium nitrate – $Ca(NO_3)_2$
• Ammonium nitrate – NH_4NO_3
• Lead(II) nitrate – $Pb(NO_3)_2$
• Copper(II) nitrate – $Cu(NO_3)_2$

Carbonates and bicarbonates
• Sodium carbonate – Na_2CO_3
• Sodium bicarbonate – $NaHCO_3$
• Potassium carbonate – K_2CO_3
• Calcium carbonate – $CaCO_3$
• Magnesium carbonate – $MgCO_3$
• Ammonium carbonate – $(NH_4)_2CO_3$

Phosphates
• Trisodium phosphate – Na_3PO_4
• Calcium phosphate – $Ca_3(PO_4)_2$
• Potassium phosphate – K_3PO_4
• Ammonium phosphate – $(NH_4)_3PO_4$
• Sodium dihydrogen phosphate – NaH_2PO_4

Other common salts
• Sodium hydroxide (base salt) – $NaOH$
• Potassium permanganate – $KMnO_4$
• Potassium dichromate – $K_2Cr_2O_7$
• Sodium thiosulfate – $Na_2S_2O_3$
• Sodium acetate – CH_3COONa
• Sodium fluoride – NaF
• Sodium bromide – $NaBr$
• Potassium iodide – KI
• Sodium sulfite – Na_2SO_3
• Sodium nitrite – $NaNO_2$
• Sodium hypochlorite – $NaClO$
• Potassium chlorate – $KClO_3$
• Sodium silicate – Na_2SiO_3
• Borax – $Na_2B_4O_7 \cdot 10H_2O$
• Potassium alum – $KAl(SO_4)_2 \cdot 12H_2O$
• Lead(II) iodide – PbI_2
• Calcium fluoride – CaF_2
• Sodium cyanide – $NaCN$

Common acid solutions
• Hydrochloric acid – HCl
• Sulfuric acid – H_2SO_4
• Nitric acid – HNO_3
• Phosphoric acid – H_3PO_4
• Acetic acid – CH_3COOH
• Carbonic acid – H_2CO_3
• Hydrofluoric acid – HF

• Citric acid – $C_6H_8O_7$

Common base solutions
• Sodium hydroxide – NaOH
• Potassium hydroxide – KOH
• Calcium hydroxide (limewater) – $Ca(OH)_2$
• Ammonia solution – NH_3(aq)
• Magnesium hydroxide – $Mg(OH)_2$

Other common solutions
• Saltwater (brine) – NaCl(aq)
• Sugar solution – $C_{12}H_{22}O_{11}$(aq)
• Hydrogen peroxide solution – H_2O_2
• Bleach – NaClO(aq)
• Vinegar – CH_3COOH(aq)
• Fehling's solution – $CuSO_4$ + sodium potassium tartrate
• Benedict's solution – $CuSO_4$ + Na_2CO_3 + sodium citrate
• Tollens' reagent – $[Ag(NH_3)_2]^+$

Metals

Alkali Metals
• Lithium – Li
• Sodium – Na
• Potassium – K
• Rubidium – Rb
• Caesium – Cs
• Francium – Fr

Alkaline Earth Metals
• Beryllium – Be
• Magnesium – Mg
• Calcium – Ca
• Strontium – Sr
• Barium – Ba
• Radium – Ra

Transition Metals
• Scandium – Sc
• Titanium – Ti
• Vanadium – V
• Chromium – Cr
• Manganese – Mn
• Iron – Fe

• Cobalt – Co
• Nickel – Ni
• Copper – Cu
• Zinc – Zn
• Yttrium – Y
• Zirconium – Zr
• Niobium – Nb
• Molybdenum – Mo
• Technetium – Tc
• Ruthenium – Ru
• Rhodium – Rh
• Palladium – Pd
• Silver – Ag
• Cadmium – Cd
• Hafnium – Hf
• Tantalum – Ta
• Tungsten – W
• Rhenium – Re
• Osmium – Os
• Iridium – Ir
• Platinum – Pt
• Gold – Au
• Mercury – Hg
• Rutherfordium – Rf
• Dubnium – Db
• Seaborgium – Sg
• Bohrium – Bh
• Hassium – Hs
• Meitnerium – Mt
• Darmstadtium – Ds
• Roentgenium – Rg
• Copernicium – Cn

Post-Transition Metals
• Aluminium – Al
• Gallium – Ga
• Indium – In
• Tin – Sn
• Thallium – Tl
• Lead – Pb
• Bismuth – Bi
• Polonium – Po
• Nihonium – Nh
• Flerovium – Fl
• Moscovium – Mc
• Livermorium – Lv

Lanthanides

• Lanthanum – La
• Cerium – Ce
• Praseodymium – Pr
• Neodymium – Nd
• Promethium – Pm
• Samarium – Sm
• Europium – Eu
• Gadolinium – Gd
• Terbium – Tb
• Dysprosium – Dy
• Holmium – Ho
• Erbium – Er
• Thulium – Tm
• Ytterbium – Yb
• Lutetium – Lu

Actinides
• Actinium – Ac
• Thorium – Th
• Protactinium – Pa
• Uranium – U
• Neptunium – Np
• Plutonium – Pu
• Americium – Am
• Curium – Cm
• Berkelium – Bk
• Californium – Cf
• Einsteinium – Es
• Fermium – Fm
• Mendelevium – Md
• Nobelium – No
• Lawrencium – Lr

Oxidizers

Elements and halogens
• Oxygen – O_2
• Ozone – O_3
• Fluorine – F_2
• Chlorine – Cl_2
• Bromine – Br_2
• Iodine – I_2

Peroxides and peroxy compounds
• Hydrogen peroxide – H_2O_2

- Sodium peroxide – Na_2O_2
- Potassium peroxide – K_2O_2
- Lithium peroxide – Li_2O_2
- Calcium peroxide – CaO_2
- Barium peroxide – BaO_2
- Magnesium peroxide – MgO_2
- Zinc peroxide – ZnO_2
- Potassium superoxide – KO_2
- Sodium percarbonate – $2Na_2CO_3 \cdot 3H_2O_2$
- Sodium perborate – $NaBO_3$
- Urea hydrogen peroxide – $CO(NH_2)_2 \cdot H_2O_2$
- Benzoyl peroxide – $(C_6H_5CO)_2O_2$
- Peracetic acid – CH_3CO_3H
- m-Chloroperoxybenzoic acid (mCPBA) – $C_7H_5ClO_3$
- tert-Butyl hydroperoxide – $(CH_3)_3COOH$
- Cumene hydroperoxide – $C_9H_{12}O_2$
- Dimethyldioxirane (DMDO) – $C_3H_6O_2$
- Caro's acid (peroxymonosulfuric acid) – H_2SO_5
- Peroxydisulfuric acid – $H_2S_2O_8$
- Ammonium persulfate – $(NH_4)_2S_2O_8$
- Sodium persulfate – $Na_2S_2O_8$
- Potassium persulfate – $K_2S_2O_8$
- Oxone – $2KHSO_5 \cdot KHSO_4 \cdot K_2SO_4$

Chlorine oxy-compounds
- Hypochlorous acid – $HClO$
- Sodium hypochlorite – $NaClO$
- Calcium hypochlorite – $Ca(ClO)_2$
- Chlorous acid – $HClO_2$
- Sodium chlorite – $NaClO_2$
- Chlorine dioxide – ClO_2
- Dichlorine monoxide – Cl_2O
- Chloric acid – $HClO_3$
- Sodium chlorate – $NaClO_3$
- Potassium chlorate – $KClO_3$
- Barium chlorate – $Ba(ClO_3)_2$
- Perchloric acid – $HClO_4$
- Sodium perchlorate – $NaClO_4$
- Potassium perchlorate – $KClO_4$
- Ammonium perchlorate – NH_4ClO_4
- Magnesium perchlorate – $Mg(ClO_4)_2$
- Dichlorine heptoxide – Cl_2O_7

Bromine and iodine oxy-compounds
- Sodium hypobromite – $NaBrO$
- Sodium bromate – $NaBrO_3$
- Potassium bromate – $KBrO_3$
- Potassium iodate – KIO_3

• Sodium periodate – $NaIO_4$
• Potassium periodate – KIO_4
• Periodic acid – HIO_4
• Iodine pentoxide – I_2O_5

Organic halogen-based oxidants
• N-Chlorosuccinimide (NCS) – $C_4H_4ClNO_2$
• N-Bromosuccinimide (NBS) – $C_4H_4BrNO_2$
• Trichloroisocyanuric acid – $C_3Cl_3N_3O_3$
• Sodium dichloroisocyanurate – $C_3Cl_2N_3NaO_3$
• Chloramine-T – $C_7H_7ClNNaO_2S$
• Dess–Martin periodinane – $C_{13}H_{13}IO_8$
• 2-Iodoxybenzoic acid (IBX) – $C_7H_5IO_4$
• Iodosobenzene – C_6H_5IO
• Selectfluor – $C_7H_{14}B_2ClF_9N_2$

Nitrogen compounds
• Nitric acid – HNO_3
• Sodium nitrate – $NaNO_3$
• Potassium nitrate – KNO_3
• Ammonium nitrate – NH_4NO_3
• Calcium nitrate – $Ca(NO_3)_2$
• Barium nitrate – $Ba(NO_3)_2$
• Strontium nitrate – $Sr(NO_3)_2$
• Lithium nitrate – $LiNO_3$
• Silver nitrate – $AgNO_3$
• Nitrogen dioxide – NO_2
• Dinitrogen tetroxide – N_2O_4
• Dinitrogen pentoxide – N_2O_5
• Nitrous oxide – N_2O
• Nitric oxide – NO
• Sodium nitrite – $NaNO_2$

Chromium compounds
• Chromium trioxide – CrO_3
• Potassium dichromate – $K_2Cr_2O_7$
• Sodium dichromate – $Na_2Cr_2O_7$
• Potassium chromate – K_2CrO_4
• Chromyl chloride – CrO_2Cl_2
• Pyridinium chlorochromate (PCC) – $C_5H_6ClCrNO_3$
• Pyridinium dichromate (PDC) – $(C_5H_5NH)_2Cr_2O_7$

Manganese compounds
• Potassium permanganate – $KMnO_4$
• Sodium permanganate – $NaMnO_4$
• Potassium manganate – K_2MnO_4
• Manganese dioxide – MnO_2
• Manganese(III) acetate – $Mn(CH_3COO)_3$

Other metal-based oxidizers
• Osmium tetroxide – OsO_4
• Ruthenium tetroxide – RuO_4
• Vanadium pentoxide – V_2O_5
• Lead dioxide – PbO_2
• Red lead – Pb_3O_4
• Lead tetraacetate – $Pb(CH_3COO)_4$
• Sodium bismuthate – $NaBiO_3$
• Silver(I) oxide – Ag_2O
• Silver(II) oxide – AgO
• Mercury(II) oxide – HgO
• Copper(II) oxide – CuO
• Copper(II) sulfate – $CuSO_4$
• Copper(II) chloride – $CuCl_2$
• Iron(III) oxide – Fe_2O_3
• Iron(III) chloride – $FeCl_3$
• Potassium ferrate – K_2FeO_4
• Potassium ferricyanide – $K_3[Fe(CN)_6]$
• Cerium(IV) ammonium nitrate – $(NH_4)_2Ce(NO_3)_6$
• Cerium(IV) sulfate – $Ce(SO_4)_2$
• Cobalt(III) fluoride – CoF_3
• Thallium(III) nitrate – $Tl(NO_3)_3$
• Gold(III) chloride – $AuCl_3$
• Mercury(II) chloride – $HgCl_2$
• Tollens' reagent – $[Ag(NH_3)_2]OH$

Sulfur, selenium, and other nonmetal oxidizers
• Sulfuric acid (concentrated) – H_2SO_4
• Sulfur trioxide – SO_3
• Selenium dioxide – SeO_2
• Selenic acid – H_2SeO_4
• Telluric acid – H_6TeO_6
• Arsenic pentoxide – As_2O_5

Fluorine-based oxidizers
• Oxygen difluoride – OF_2
• Dioxygen difluoride – O_2F_2
• Chlorine trifluoride – ClF_3
• Bromine trifluoride – BrF_3
• Iodine pentafluoride – IF_5
• Xenon difluoride – XeF_2
• Xenon trioxide – XeO_3
• Antimony pentafluoride – SbF_5

Organic quinone and radical oxidants
• p-Benzoquinone – $C_6H_4O_2$
• Chloranil – $C_6Cl_4O_2$

• DDQ (2,3-dichloro-5,6-dicyano-1,4-benzoquinone) – $C_8Cl_2N_2O_2$
• Fremy's salt – $K_2[ON(SO_3)_2]$

Indicators

Acid-Base (pH) Indicators
• Methyl violet: $C_{24}H_{28}ClN_3$
• Crystal violet: $C_{25}H_{30}ClN_3$
• Malachite green: $C_{23}H_{25}ClN_2$
• Thymol blue: $C_{27}H_{30}O_5S$
• p-Xylenol blue: $C_{29}H_{32}O_5S$
• Methyl yellow: $C_{14}H_{15}N_3$
• Bromophenol blue: $C_{19}H_{10}Br_4O_5S$
• Bromochlorophenol blue: $C_{19}H_{10}Br_2Cl_2O_5S$
• Congo red: $C_{32}H_{22}N_6Na_2O_6S_2$
• Methyl orange: $C_{14}H_{14}N_3NaO_3S$
• Tropaeolin OO (Orange IV): $C_{18}H_{14}N_3NaO_3S$
• Tropaeolin O: $C_{12}H_9N_2NaO_5S$
• Bromocresol green: $C_{21}H_{14}Br_4O_5S$
• Methyl red: $C_{15}H_{15}N_3O_2$
• Chlorophenol red: $C_{19}H_{12}Cl_2O_5S$
• Bromocresol purple: $C_{21}H_{16}Br_2O_5S$
• Bromothymol blue: $C_{27}H_{28}Br_2O_5S$
• Phenol red: $C_{19}H_{14}O_5S$
• Cresol red: $C_{21}H_{18}O_5S$
• Metacresol purple: $C_{21}H_{18}O_5S$
• Neutral red: $C_{15}H_{17}ClN_4$
• Phenolphthalein: $C_{20}H_{14}O_4$
• Thymolphthalein: $C_{28}H_{30}O_4$
• o-Cresolphthalein: $C_{22}H_{18}O_4$
• α-Naphtholphthalein: $C_{28}H_{18}O_4$
• Alizarin yellow R: $C_{13}H_8N_3NaO_5$
• Alizarin: $C_{14}H_8O_4$
• Alizarin red S: $C_{14}H_7NaO_7S$
• Indigo carmine: $C_{16}H_8N_2Na_2O_8S_2$
• Quinaldine red: $C_{21}H_{23}IN_2$
• Brilliant yellow: $C_{26}H_{18}N_4Na_2O_8S_2$
• Titan yellow: $C_{28}H_{19}N_5Na_2O_6S_4$
• 2,4-Dinitrophenol: $C_6H_4N_2O_5$
• 4-Nitrophenol (p-Nitrophenol): $C_6H_5NO_3$
• Litmus (azolitmin): ~$C_7H_7NO_4$
• Cyanidin (anthocyanin): $C_{15}H_{11}O_6^+$
• Curcumin (turmeric): $C_{21}H_{20}O_6$

• Universal indicator: mixture of several dyes

Redox Indicators
• Methylene blue: $C_{16}H_{18}ClN_3S$
• Thionine: $C_{12}H_{10}ClN_3S$
• Safranin O: $C_{20}H_{19}ClN_4$
• Phenosafranine: $C_{18}H_{15}ClN_4$
• Janus green B: $C_{30}H_{31}ClN_6$
• Ferroin: $[Fe(C_{12}H_8N_2)_3]^{2+}$
• Nitroferroin: $[Fe(C_{12}H_7N_3O_2)_3]^{2+}$
• Diphenylamine: $C_{12}H_{11}N$
• Sodium diphenylamine sulfonate: $C_{12}H_{10}NNaO_3S$
• N-Phenylanthranilic acid: $C_{13}H_{11}NO_2$
• DCPIP (2,6-dichlorophenolindophenol): $C_{12}H_7Cl_2NO_2$
• Resazurin: $C_{12}H_7NO_4$
• Methyl viologen: $C_{12}H_{14}Cl_2N_2$
• Starch (iodine indicator): $(C_6H_{10}O_5)$□
• Potassium permanganate (self-indicator): $KMnO_4$

Complexometric (Metal Ion) Indicators
• Eriochrome Black T: $C_{20}H_{12}N_3NaO_7S$
• Eriochrome Blue-Black R: $C_{20}H_{13}N_2NaO_5S$
• Calmagite: $C_{17}H_{14}N_2O_5S$
• Murexide: $C_{88}H_8N_6O_6$ → correctly **$C_8H_8N_6O_6$**
• Xylenol orange: $C_{31}H_{32}N_2O_{13}S$
• Methylthymol blue: $C_{37}H_{43}N_2NaO_{13}S$
• Calcein: $C_{30}H_{26}N_2O_{13}$
• Pyrocatechol violet: $C_{19}H_{14}O_7S$
• PAN (1-(2-pyridylazo)-2-naphthol): $C_{15}H_{11}N_3O$
• PAR (4-(2-pyridylazo)resorcinol): $C_{11}H_9N_3O_2$
• Zincon: $C_{20}H_{16}N_4O_6S$
• Dithizone: $C_{13}H_{12}N_4S$
• Diphenylcarbazone: $C_{13}H_{12}N_4O$
• Sulfosalicylic acid: $C_7H_6O_6S$
• Salicylic acid: $C_7H_6O_3$
• 1,10-Phenanthroline: $C_{12}H_8N_2$
• Dimethylglyoxime: $C_4H_8N_2O_2$
• Tiron: $C_6H_4Na_2O_8S_2$
• Chrome azurol S: $C_{23}H_{13}Cl_2Na_3O_9S$

Precipitation / Adsorption Indicators
• Potassium chromate (Mohr): K_2CrO_4
• Ferric ammonium sulfate (Volhard): $NH_4Fe(SO_4)_2$
• Fluorescein: $C_{20}H_{12}O_5$
• Dichlorofluorescein: $C_{20}H_{10}Cl_2O_5$
• Eosin Y: $C_{20}H_6Br_4Na_2O_5$
• Rhodamine 6G: $C_{28}H_{31}ClN_2O_3$

Fluorescent Indicators
• Quinine: $C_{20}H_{24}N_2O_2$
• Acridine: $C_{13}H_9N$
• Acridine orange: $C_{17}H_{19}N_3$
• Umbelliferone: $C_9H_6O_3$
• 4-Methylumbelliferone: $C_{10}H_8O_3$
• Rhodamine B: $C_{28}H_{31}ClN_2O_3$

Catalysts

Heterogeneous (metals and oxides)
- Iron: Fe
- Platinum: Pt
- Palladium: Pd
- Rhodium: Rh
- Nickel: Ni
- Raney nickel: Ni-Al
- Copper: Cu
- Silver: Ag
- Gold: Au
- Ruthenium: Ru
- Iridium: Ir
- Cobalt: Co
- Vanadium(V) oxide: V_2O_5
- Manganese dioxide: MnO_2
- Titanium dioxide: TiO_2
- Aluminium oxide: Al_2O_3
- Zinc oxide: ZnO
- Iron(III) oxide: Fe_2O_3
- Cerium(IV) oxide: CeO_2
- Molybdenum disulfide: MoS_2
- Copper–zinc oxide/alumina: $Cu/ZnO/Al_2O_3$
- Palladium on carbon: Pd/C
- Platinum on carbon: Pt/C
- Lindlar catalyst: $Pd/CaCO_3$, Pb-poisoned

Homogeneous (metal complexes)
- Wilkinson's catalyst: $RhCl(PPh_3)_3$
- Grubbs' catalyst (1st generation): $RuCl_2(=CHPh)(PCy_3)_2$
- Grubbs' catalyst (2nd generation): $RuCl_2(=CHPh)(IMes)(PCy_3)$
- Schrock catalyst: $Mo(=NAr)(=CHCMe_2Ph)(OR)_2$
- Crabtree's catalyst: $[Ir(cod)(PCy_3)(py)]PF_6$
- Vaska's complex: $IrCl(CO)(PPh_3)_2$
- Ziegler–Natta catalyst: $TiCl_4/Al(C_2H_5)_3$
- Zeise's salt: $K[PtCl_3(C_2H_4)]$

- Tetrakis(triphenylphosphine)palladium(0): $Pd(PPh_3)_4$
- Palladium(II) acetate: $Pd(OAc)_2$
- Palladium(II) chloride: $PdCl_2$
- Wacker catalyst: $PdCl_2/CuCl_2$
- Hoveyda–Grubbs catalyst: $RuCl_2(=CH\text{-}C_6H_4\text{-}O\text{-}iPr)(IMes)$
- Jacobsen's catalyst: Mn(salen)Cl
- Sharpless catalyst: $Ti(OiPr)_4$/diethyl tartrate
- Noyori catalyst: $RuCl_2(BINAP)$
- Cobalt carbonyl: $Co_2(CO)_8$
- Rhodium(III) chloride: $RhCl_3$
- Monsanto catalyst: $[Rh(CO)_2I_2]^-$
- Cativa catalyst: $[Ir(CO)_2I_2]^-$
- Ferrocene-based catalysts: $Fe(C_5H_5)_2$
- Kaminsky catalyst: Cp_2ZrCl_2/MAO

Acid and base catalysts
- Sulfuric acid: H_2SO_4
- Hydrochloric acid: HCl
- Phosphoric acid: H_3PO_4
- Nitric acid: HNO_3
- p-Toluenesulfonic acid: $CH_3C_6H_4SO_3H$
- Aluminium chloride: $AlCl_3$
- Boron trifluoride: BF_3
- Iron(III) chloride: $FeCl_3$
- Zinc chloride: $ZnCl_2$
- Titanium tetrachloride: $TiCl_4$
- Sodium hydroxide: NaOH
- Potassium hydroxide: KOH
- Sodium ethoxide: $NaOC_2H_5$
- Potassium tert-butoxide: $KOC(CH_3)_3$
- Triethylamine: $N(C_2H_5)_3$
- Pyridine: C_5H_5N
- DMAP: $(CH_3)_2NC_5H_4N$
- DBU: $C_9H_{16}N_2$
- Zeolite ZSM-5: $Na□[Al□Si_{96}\text{-}□O_{192}]\cdot 16H_2O$
- Zeolite Y: $Na_{56}[Al_{56}Si_{136}O_{384}]\cdot 250H_2O$
- Amberlyst-15: sulfonated polystyrene resin

Organocatalysts
- Proline: $C_5H_9NO_2$
- Thiourea: CH_4N_2S
- N-heterocyclic carbenes: e.g. IMes, $C_{21}H_{24}N_2$
- Cinchona alkaloids: e.g. quinine, $C_{20}H_{24}N_2O_2$
- MacMillan catalyst: $C_{14}H_{20}N_2O$
- Jørgensen–Hayashi catalyst: $C_{21}H_{27}NOSi$

Phase-transfer and others
- Tetrabutylammonium bromide: $(C_4H_9)_4NBr$

- 18-Crown-6: $C_{12}H_{24}O_6$
- Benzyltriethylammonium chloride: $C_{13}H_{22}NCl$

Biological (enzymes) and industrial named catalysts
- Catalase: Fe-heme enzyme
- Carbonic anhydrase: Zn-containing enzyme
- Nitrogenase: Fe/Mo-cofactor (FeMo-co): $MoFe_7S_9C$
- Haber–Bosch catalyst: $Fe/K_2O/Al_2O_3$
- Ostwald process catalyst: Pt/Rh gauze
- Contact process catalyst: V_2O_5
- Three-way automotive catalyst: Pt/Pd/Rh on CeO_2/Al_2O_3
- Fischer–Tropsch catalysts: Fe or Co
- Hydrodesulfurization catalyst: $CoMoS/Al_2O_3$ or $NiMoS/Al_2O_3$
- Methanol synthesis catalyst: $Cu/ZnO/Al_2O_3$
- Steam reforming catalyst: Ni/Al_2O_3